

Vaghesan Sundaram

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Education

University of Maryland, College Park

B.S. Computer Science, Cybersecurity Focus; B.S. Mathematics; ACES Cybersecurity Minor

Expected Fall 2027

College Park, MD

Relevant Coursework: Data Structures, Algorithms, Computer Systems, Statistics, Cryptography, Network Security, HACS408T Penetration Testing

Middlesex College

A.S. Mathematics

June 2024

Edison, NJ

Technical Skills

Languages: Python, Java, C++

Technologies: Flask, SQLite, NumPy, SciPy, scikit-learn, Git, Linux, PowerShell, REST APIs, JSON/JSON Schema, automated testing

Selected Projects

Agent Least Privilege

GitHub

- Built a 60-cell synthetic tool-use evaluation with gpt-5.4-mini; a task-conditioned monitor prevented all observed unauthorized state-changing effects (0/10) versus 4/10 unrestricted and 6/10 with a static allowlist.
- Separated model proposals from effects: the monitored model still proposed unauthorized actions in 5/10 attack cases, but the external monitor blocked each observed effect.

Keystroke Authentication System

GitHub

- Built a two-service Flask authentication system with SQLite persistence, typing-cadence validation, JWT-bound flows, bcrypt fallback, rate limiting, and single-use server-to-server code exchange.
- Added an end-to-end smoke test covering account creation, three-sample enrollment, and authentication.

Prophet Prediction Market

GitHub

- Built a Python event-to-market pipeline that normalized social posts, weighted signals by recency and engagement, clustered sentence embeddings, and generated candidate markets.
- Defined structured WebSocket create/resolve actions that integrated the engine with independently developed exchange, agent, and on-chain services at HackPrinceton 2025.

Experience

Cybersecurity Intern

Jun 2026 to Present

Redmind Technologies

Remote

- Assessed one public-facing client site with Semgrep and manual review; documented 7 findings across authentication, data exposure, configuration, and dependency risk.
- Mapped evidence and severity decisions to remediation guidance, then presented a written report and slide deck to the developer team.

Information Technology Intern

Jun to Aug 2025

New Jersey Courts

Toms River, NJ

- Supported production IT systems across court departments through troubleshooting, access workflows, user support, documentation, and escalation.
- Documented recurring issues and support procedures for repeatable troubleshooting and clearer team handoffs.

Certification

CompTIA Security+ (SY0-701), issued March 2026; expires March 2029.